AURAASSESS: BEHAVIOR & INTERVIEW OUTCOMES ANALYTICS

A Multimodal AI Psychological Inference Engine

1. Technology Stack Used

AuraAssess utilizes a premium multimodal late fusion architecture, capturing high-frequency video and audio inputs to infer candidate behavioral profiles. Below is a detailed breakdown of the technical stack:

Category	Technologies & Libraries	Purpose & Applications
Computer Vision	MediaPipe Face Mesh, MediaPipe Pose, MediaPipe Iris, OpenCV	Extracts eyebrow furrowing AU4, raise AU1/2, blink EAR, gaze deviation, postural sway, lean angle, crossed arms, and self-touches.
Signal & Audio	Librosa, sounddevice, soundfile, Whisper (VAD)	Records real-time microphone stream, computes voice pitch F0 variance, voice energy variance, and speech response latency.
Machine Learning	scikit-learn, joblib, NumPy, Pandas, SciPy	Runs session-level train-test splits. Trains Random Forest Classifier on Z-scored features. Includes mathematical Z-score engine.
Web & Interface	FastAPI, Uvicorn, WebSockets, HTML5, CSS3	Ingests camera frames from client browsers in real-time, displays live predictions, overlays landmark mesh, and renders executive report.

2. System Execution & Architecture

AuraAssess is executed in two core operational modes: the **Real-Time Web Dashboard** and the **Standalone Command-Line Runner**.

Real-Time Web Dashboard (server.py & app.js)

- Client webcam feeds frames via WebSocket connection at 10 FPS (100ms intervals) to the FastAPI server.
  - The server processes each frame through MediaPipe and OpenCV, returning landmark coordinate arrays.
  - Feature aggregation runs on-the-fly, generating behavioral outcomes and timeline reports inside the client browser.

Standalone Command-Line Runner (main.py)

- Spawns concurrent background threads: `VideoCaptureThread` for camera/video processing and `AudioCaptureThread` for sound recording.
  - Audio processing computes vocal energy variance and uses YIN to extract voiced pitch frequencies (F_0).
  - A speech latency tracker measures the response gap between question endings and candidate speaking cues.

Calibration & Baseline Mechanism

To mitigate neurodiversity and cultural bias, the system begins with an active **15-second Calibration Phase**. During this window, baseline means and standard deviations are computed for each parameter: $$z(t) = \frac{x(t) - \mu_{baseline}}{\sigma_{baseline}}$$
Subsequent observations are evaluated as normalized deviations (Z-scores) from the candidate's custom personal baseline.

3. Pros & Cons Analysis

AuraAssess represents a highly adaptive approach to candidate assessment, but like all AI models, it has specific trade-offs:

System Strengths (Pros)	Limitations & Challenges (Cons)
• **Bias Mitigation:** In-session calibration compares the user only against their own neutral baseline, avoiding rigid global benchmarks.	• **High Computational Load:** Real-time frame processing of Face Mesh and Pose models requires considerable local CPU/GPU capability.
• **Non-Intrusive:** Requires only standard webcam and microphone hardware; no special bio-sensors or expensive setups needed.	• **Audio Extraction Complexity:** Real-time pitch tracking (F_0) is highly sensitive to background noise and acoustic echo.
• **Comprehensive Kinesics:** Tracks advanced somatic behaviors (crossed arms, hand fidgets, postural sway, chest distance).	• **Camera Position Dependency:** Posture lean and facial geometry indexes rely on a stable, front-facing camera angle.
• **Real-Time Outputs:** Operates with edge latency below 150ms, facilitating live dashboards and quick assessments.	• **Privacy Constraints:** Recording and processing feeds requires strict client consent and explicit data isolation.

4. Use Case Scenarios

AuraAssess applies behavioral and outcome predictions across multiple fields, from hiring to training:

Corporate Recruitment & HR Screening

During large-scale virtual interviews, HR departments use the executive summary report to analyze candidate confidence, engagement levels, and anxiety spikes under technical questions. The Z-score engine ensures candidates with naturally quiet voices or different baseline mannerisms are evaluated fairly, measuring deviation rather than comparisons to a rigid norm.

Professional Communication & Speaking Coach

Public speakers, executives, and sales representatives use the dashboard tool as an interactive coaching portal. By reviewing their timeline data, users learn when they exhibit pacifying hand movements, crossed arms, flat vocal delivery, or high postural sway (fidgeting), enabling targeted self-correction.

Interview Prep & Neurodivergent Training Tools

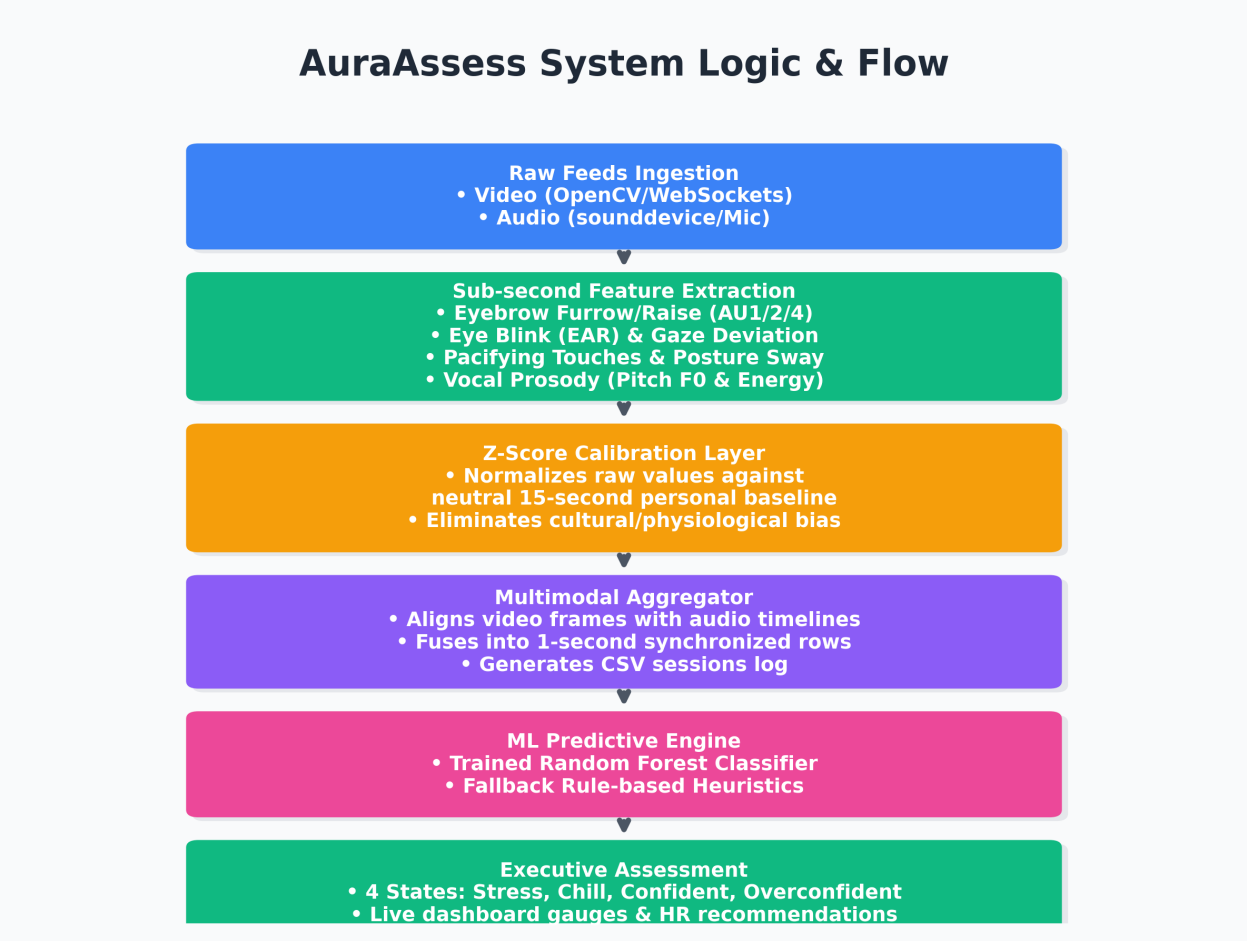
Job seekers use the dashboard to practice mock interviews. For neurodivergent individuals, it acts as a helpful biofeedback interface, teaching how posture leans, hand gesture expressiveness, and speech pacing correspond to perceived confidence (Confident vs. Stress or Overconfident) in standard interview environments.

Educational Mock Assessments

Universities utilize the command-line tool (`main.py`) to run automated mock placement interviews, helping students review speaking latency, prosodic modulation, and posture composure under pressure.

5. Diagram: System Workflow & Logic

The following diagram illustrates the flow of raw input streams through the feature extraction layers, baseline calibration, multimodal second-by-second aggregation, and the machine learning classification block:



6. Diagram: Technology Architecture

The diagram below outlines the logical layering of technology in AuraAssess, showing how data flows from ingestion hardware up through processing pipelines to the user dashboard:

